HOSPITAL BILLING AND PATIENT CLASSIFICATION

Python Programming Case Study

1. Problem Statement

Hospitals need an accurate way to calculate patient bills using consultation, diagnostic, and treatment charges. Manual calculations may cause errors when applying concessions and classifying payments. A program is needed to validate charges, calculate the final bill, and assign a payment category.

2. Objective

To develop a Python program that accepts patient details and hospital charges, validates billing constraints, applies eligible concessions, calculates the final bill, and classifies the payment.

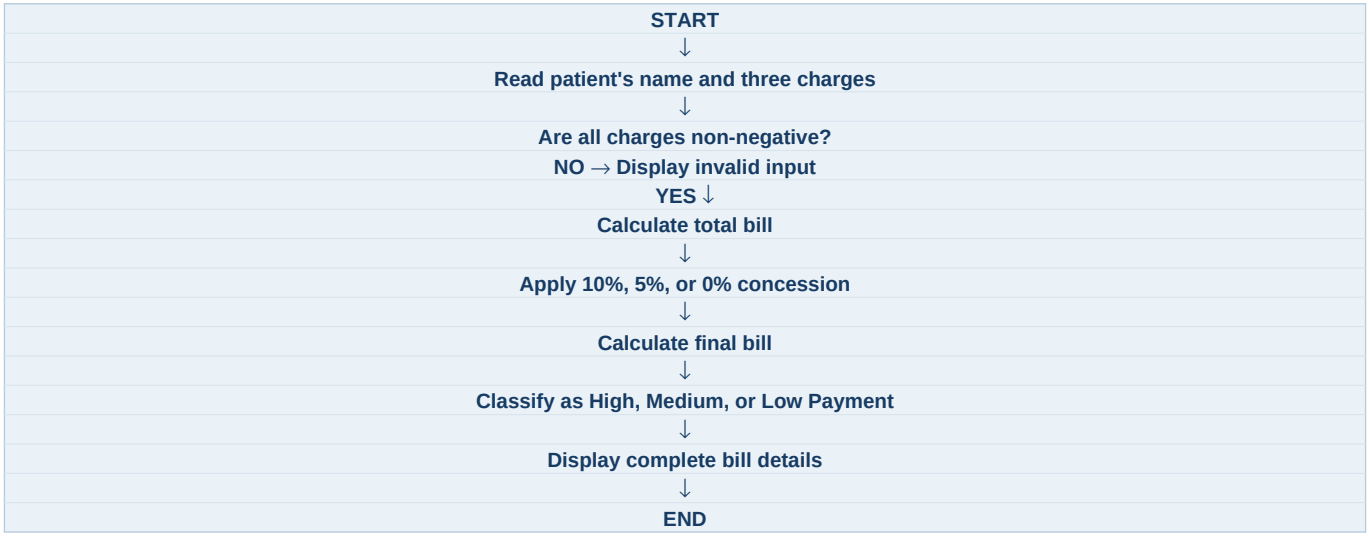
3. Assumptions

- All consultation, diagnostic, and treatment charges must be numeric and non-negative.
- The total bill is the sum of consultation, diagnostic, and treatment charges.
- A 10% concession is applied when the total bill is INR 10,000 or more.
- A 5% concession is applied when the total bill is from INR 5,000 to below INR 10,000.
- No concession is applied when the total bill is below INR 5,000.
- The concession is calculated on the total bill and subtracted to get the final bill.
- Payment classification: High Payment for a final bill of INR 10,000 or more; Medium Payment for INR 5,000 or more but below INR 10,000; otherwise Low Payment.
- Invalid negative charges are rejected and the user is asked to enter valid values.

4. Algorithm

1. Start the program and read the patient's name.
2. Read consultation, diagnostic, and treatment charges.
3. Validate that all charges are numeric and non-negative; if invalid, request valid input.
4. Add the three charges to calculate the total bill.
5. If the total is INR 10,000 or more, calculate a 10% concession.
6. Else if the total is INR 5,000 or more, calculate a 5% concession; otherwise concession is zero.
7. Subtract the concession from the total to calculate the final bill.
8. Classify the patient as High, Medium, or Low Payment according to the final bill.
9. Display the bill details and stop.

5. Flowchart



6. Pseudocode

```
BEGIN
    INPUT patient_name
    INPUT consultation, diagnostic, treatment

    IF consultation < 0 OR diagnostic < 0 OR treatment < 0
        DISPLAY "Invalid charges"
        STOP
    END IF

    total = consultation + diagnostic + treatment

    IF total >= 10000
        concession = total * 0.10
    ELSE IF total >= 5000
```

```
        concession = total * 0.05
    ELSE
        concession = 0
    END IF

    final_bill = total - concession

    IF final_bill >= 10000
        category = "High Payment"
    ELSE IF final_bill >= 5000
        category = "Medium Payment"
    ELSE
        category = "Low Payment"
    END IF

    DISPLAY patient_name, total, concession
    DISPLAY final_bill, category
END
```

7. Python Code

```
name = input("Enter patient name: ")

consultation = float(input("Enter consultation charges: "))
diagnostic = float(input("Enter diagnostic charges: "))
treatment = float(input("Enter treatment charges: "))

# Validate billing constraints
if consultation < 0 or diagnostic < 0 or treatment < 0:
    print("Invalid charges. Enter non-negative values.")
else:
    total = consultation + diagnostic + treatment

    if total >= 10000:
        concession = total * 0.10
    elif total >= 5000:
        concession = total * 0.05
    else:
        concession = 0

    final_bill = total - concession

    if final_bill >= 10000:
        category = "High Payment"
    elif final_bill >= 5000:
        category = "Medium Payment"
    else:
        category = "Low Payment"

    print("\\nHOSPITAL BILL DETAILS")
    print("Patient Name:", name)
    print("Consultation Charges:", consultation)
    print("Diagnostic Charges:", diagnostic)
    print("Treatment Charges:", treatment)
    print("Total Bill:", round(total, 2))
    print("Concession:", round(concession, 2))
    print("Final Bill:", round(final_bill, 2))
    print("Payment Category:", category)
```

8. Expected Output (Illustration)

```
Enter patient name: Rahul
Enter consultation charges: 2000
Enter diagnostic charges: 3500
Enter treatment charges: 6000

HOSPITAL BILL DETAILS
Patient Name: Rahul
Consultation Charges: 2000
Diagnostic Charges: 3500
Treatment Charges: 6000
Total Bill: 11500.0
Concession: 1150.0
Final Bill: 10350.0
Payment Category: High Payment
```